

Partial Trace: Intuition



- In many situations, one part of the system (e.g., Bob's) is inaccessible.
- We then describe Alice's state using the **partial trace**: $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$
- The result ρ_A is called the **reduced state** of Alice.
- The **partial trace** is the mathematical operation that "traces out" or "ignores" subsystem B
- This captures all the information Alice can obtain about her part of the system.



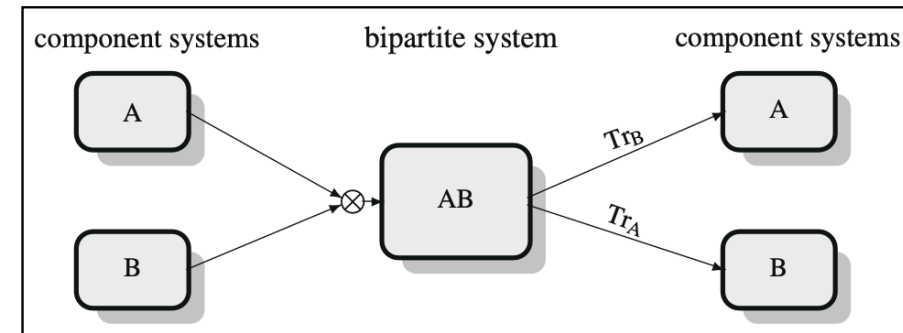
Partial Trace: Definition

- Consider a density operator ρ_{AB} acting on the combined Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$.
- Let $\{ |i_B\rangle \}_i$ be an orthonormal basis for subsystem B .
- The partial trace over B is defined as: $\text{Tr}_B(\rho_{AB}) = \sum_i (I_A \otimes \langle i_B |) \rho_{AB} (I_A \otimes |i_B\rangle)$.
- For simplicity, we usually write: $\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_i \langle i_B | \rho_{AB} | i_B \rangle$.
- The state ρ_A is the reduced density matrix of subsystem A .
- PT over B for elementary operators:** $\text{Tr}_B[|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|] = |a_1\rangle\langle a_2| \text{Tr}[|b_1\rangle\langle b_2|]$
where $\text{Tr}[|b_1\rangle\langle b_2|] = \langle b_1 | b_2 \rangle$.

- Proposition:** Let ρ be a density operator of the bipartite system $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ and let $\rho(a_1, a_2, b_1, b_2)$ be a matrix representation of ρ , where a_1, a_2 refers to a basis of $\mathcal{H}_A$ and b_1, b_2 to a basis of $\mathcal{H}_B$. Then the matrix representations of the reduced operators are given by

$$\rho_A(a_1, a_2) = \sum_b \rho(a_1, a_2, b, b), \text{ and}$$

$$\rho_B(b_1, b_2) = \sum_a \rho(a, a, b_1, b_2).$$





Properties of Reduced State

- ρ_A is a valid density operator:
 - It is **positive semi-definite**.
 - It has trace equal to 1.
- Taking partial traces in any order recovers the full trace: $\text{Tr}(\rho_{AB}) = \text{Tr}_A(\text{Tr}_B(\rho_{AB})) = \text{Tr}_B(\text{Tr}_A(\rho_{AB}))$.
- It describes the state of a subsystem when the rest of the system is ignored or inaccessible.
- Even if the overall state ρ_{AB} is **pure**, the reduced state ρ_A may be **mixed** due to entanglement.
- This tool helps analyze entanglement, decoherence, and measurement outcomes on subsystems.



Bell State Example

- Consider the Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.
- This is a pure entangled state of two qubits shared between Alice and Bob.
- Let's compute the reduced states $\rho_A = \text{Tr}_B(\rho_{AB})$ and $\rho_B = \text{Tr}_A(\rho_{AB})$.
- The full density operator is given by: $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$.
- Expand using the definition of $|\Phi^+\rangle$: $\rho_{AB} = \frac{1}{2} (|00\rangle + |11\rangle) (\langle 00| + \langle 11|)$.
- Multiply out the outer product: $\rho_{AB} = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$.



Reduced State of A: $\rho_A = \text{Tr}_B(\rho_{AB})$

- We trace out Bob's system using the basis $\{ |0\rangle_B, |1\rangle_B \}$.

- Compute each term:

- $\text{Tr}_B(|00\rangle\langle 00|) = |0\rangle_A\langle 0| \cdot \underbrace{\langle 0|0\rangle}_1 = |0\rangle\langle 0|.$

- $\text{Tr}_B(|00\rangle\langle 11|) = |0\rangle_A\langle 1| \cdot \underbrace{\langle 0|1\rangle}_0 = 0.$

- $\text{Tr}_B(|11\rangle\langle 00|) = |1\rangle_A\langle 0| \cdot \underbrace{\langle 1|0\rangle}_0 = 0.$

- $\text{Tr}_B(|11\rangle\langle 11|) = |1\rangle_A\langle 1| \cdot \underbrace{\langle 1|1\rangle}_1 = |1\rangle\langle 1|.$

- So, $\rho_A = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I_{\mathcal{H}_A}.$

- This is the **maximally mixed state**.



Reduced State of B: $\rho_B = \text{Tr}_A(\rho_{AB})$

- Now trace out Alice's subsystem using basis $\{ |0\rangle_A, |1\rangle_A \}$.

- Compute each term:

$$\bullet \text{Tr}_A(|00\rangle\langle 00|) = |0\rangle_B\langle 0| \cdot \underbrace{\langle 0|0\rangle}_1 = |0\rangle\langle 0|.$$

$$\bullet \text{Tr}_A(|00\rangle\langle 11|) = |0\rangle_B\langle 1| \cdot \underbrace{\langle 0|1\rangle}_0 = 0.$$

$$\bullet \text{Tr}_A(|11\rangle\langle 00|) = |1\rangle_B\langle 0| \cdot \underbrace{\langle 1|0\rangle}_0 = 0.$$

$$\bullet \text{Tr}_A(|11\rangle\langle 11|) = |1\rangle_B\langle 1| \cdot \underbrace{\langle 1|1\rangle}_1 = |1\rangle\langle 1|.$$

- So, $\rho_B = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I_{\mathcal{H}_B}$
- Bob also sees a **maximally mixed state**.



What Does This Tell Us?

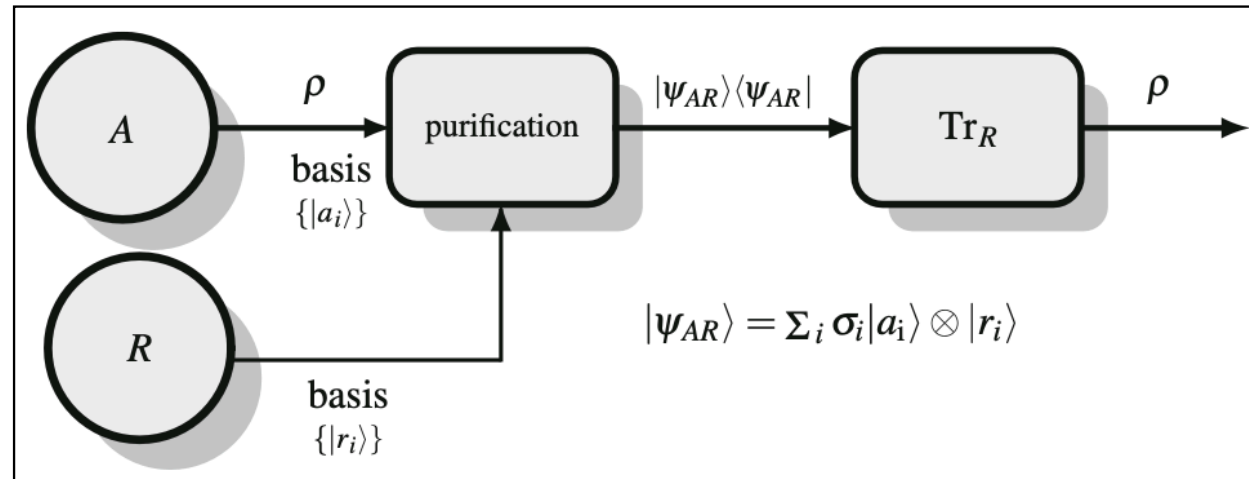
$$\text{But: } |\Phi^+\rangle_{AB}\langle\Phi^+| \neq \frac{1}{2}I_{\mathcal{H}_A} \otimes \frac{1}{2}I_{\mathcal{H}_B} !$$

- Both subsystems are in a maximally mixed state.
- But the total state ρ_{AB} is pure and entangled.
- The subsystems appear completely random if measured alone, but they are perfectly correlated when considered together.
- This is a signature of **quantum entanglement**.



Purification of Mixed States

- **Purification:** A mixed state described by density operator can be represented as a pure state provided that the quantum environment is appropriately enlarged.
- For a quantum system A with density operator ρ_A , enlargement is obtained by introducing a copy R of the system A to form the joint system AR s.t. $\rho_A = \text{Tr}_R[|\psi_{AR}\rangle\langle\psi_{AR}|]$
- **Proposition:** Let ρ_A be a density operator of A and let $\rho_A = \sum_i \sigma_i^2 |a_i\rangle\langle a_i|$ be its eigenvalue decomposition. Then a **purification** of ρ_A is given by the composite state: $|\psi_{AR}\rangle = \sum_i \sigma_i |a_i\rangle \otimes |r_i\rangle$ where $\{r_i\}$ is a basis of R .



Overview



- Quantum Information Concepts
- **Quantum Entropy**
- Quantum Data Compression
- Quantum Channels
- Quantum Capacity

Motivation: Classical vs Quantum Entropy



- Classical entropy $H(X)$ measures uncertainty in a random variable based on its probability distribution.
- In the quantum setting, uncertainty arises both from classical randomness and quantum uncertainty.
- This is captured by the **density operator** ρ , which describes the quantum state.
- The quantum analogue of Shannon entropy is the **von Neumann entropy**.



Definition of von Neumann Entropy

- Let $\rho_A \in \mathcal{B}(\mathcal{H}_A)$ be a density operator of quantum system A .
- **Von Neumann Entropy:** $S(A)_\rho = -\text{Tr}(\rho_A \log \rho_A)$
- **Notation:** $S(A)_\rho \equiv S(\rho_A) \equiv S(A)$
- If ρ_A has spectral decomposition $\rho_A = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$, then:

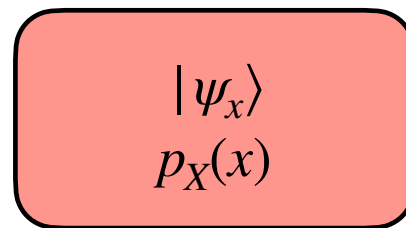
$$S(\rho_A) = - \sum_j \lambda_j \log \lambda_j$$

- The entropy is the function of eigenvalues λ_j of ρ .
- The eigenvalues form a probability distribution.
- Quantum entropy quantifies uncertainty in a quantum state, similar to classical entropy.

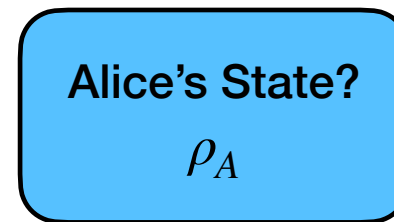


Interpretation and Examples

- **Example:** If Alice prepares a quantum state $|\psi_x\rangle$ with probability $p_X(x)$, the density matrix is:
$$\rho_A = \sum_x p_X(x) |\psi_x\rangle\langle\psi_x|$$
- The entropy $S(A)_\rho$ captures how much information Bob gains upon learning the state.



Alice



Bob



Properties of Quantum Entropy

- **Property-1 Entropy of Pure States:** The quantum entropy $S(\rho)$ vanishes iff ρ is a pure state.
 - If $S(\rho) = 0$, then $\lambda_j \log \lambda_j = 0 \forall j \implies \lambda_j = 0$ or $\lambda_j = 1$. Since $\sum_j \lambda_j = 1$, exactly one $\lambda_{j^*} = 1$.
 - If ρ is pure : $\rho = |\psi\rangle\langle\psi|$, then $S(\rho) = -\log(1) = 0$.
- **Property-2 Non-negativity of Entropy:** The quantum entropy $S(\rho)$ is always non-negative $S(\rho) \geq 0$
 - This follows from the fact that the Shannon entropy is non-negative and $\{\lambda_j\}$ form a valid probability distribution.
- **Property-3 Maximum Entropy:** The quantum entropy is bounded above by $\log(d)$, where d is the dimension of the Hilbert space: $S(\rho) \leq \log d$
 - The maximum is achieved for the maximally mixed state: $\rho = \frac{1}{d}I$.
 - Its eigenvalues are uniform: $\lambda_j = \frac{1}{d}$ for all j .
 - Then $S(\rho) = -\sum_j \frac{1}{d} \log \frac{1}{d} = \log d$.



Properties of Quantum Entropy

- **Property-4 Invariance Under Isometries:** Entropy is invariant under isometries:

- $S(\rho) = S(V\rho V^\dagger)$

$$V\rho V^\dagger = \sum_j \lambda_j V|\psi_j\rangle\langle\psi_j| V^\dagger$$

Reason:

- $= \sum_j \lambda_j |\psi'_j\rangle\langle\psi'_j|$

- The eigenvalues remain unchanged, hence entropy remains the same.

- **Property-5 Concavity of Quantum Entropy:** Quantum entropy is concave in the density operator.

- Let $\{p_X(x), \rho_x\}$ be a set of density operators and probabilities, then

$$S\left(\sum_x p_X(x)\rho_x\right) \geq \sum_x p_X(x)S(\rho_x)$$

- **Intuition:** Mixing quantum states increases uncertainty.

Quantum Entropies



- These properties enable quantum generalizations of classical quantities like conditional entropy, joint entropy, and mutual information.

**Joint
Entropy**

**Conditional
Entropy**

**Coherent
Information**

**Mutual
Information**